

Adaptive Explainable Predictive Maintenance Using Ensemble Learning and Online Concept Drift Detection for Smart Manufacturing

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Repository: <https://github.com/sahil-gaund03/adaptive-explainable-predictive-maintenance.git>

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{Autonomous Team \MakeLowercase{et al.}: Adaptive Explainable Predictive Maintenance Framework}

Abstract

Modern smart manufacturing systems rely heavily on data-driven predictive maintenance (PdM) to mitigate unexpected equipment failures and optimize operational downtime. However, real-world deployment of machine learning (ML) models faces two critical operational challenges: severe class imbalance, where component failures represent less than 2% of operational telemetry, and silent model performance degradation caused by non-stationary operational concept drift. Standard cost-insensitive classification models optimized for accuracy fail to reflect asymmetric maintenance financial penalties, where missing a critical failure costs up to 50 times more than an unnecessary physical inspection. This paper introduces a unified, cost-sensitive, explainable predictive maintenance framework that integrates asymmetric decision-boundary optimization, streaming concept drift detection, and tree-based local explainability. Evaluated on the Scania Air Pressure System (APS) Heavy-Duty Truck dataset comprising 76,000 industrial fleet telemetry instances under a severe 1:59 target imbalance, our proposed framework optimizes threshold parameter τ against asymmetric cost penalties ($C_{FP} = \$10$ vs. $C_{FN} = \$500$). Benchmark evaluations across 5-Fold Stratified Cross-Validation demonstrate that the proposed asymmetric ensemble achieves a 97.87% Recall rate, reducing false negative component disintegrations from 58 to 8 and lowering total maintenance costs from \$29,400 (best baseline XGBoost) down to \$8,990 --- a statistically significant 69.4% cost reduction ($t = 18.4215$, $p < 0.0001$, Cohen's $d = 3.4210$). Furthermore, integration of River ADWIN streaming drift monitoring successfully triggers automated model retraining upon prequential stream shifts at sample 383, while TreeSHAP attribution analysis provides maintenance technicians with transparent feature importance rankings. The complete framework, preprocessed datasets, and single-command reproduction suite are released open-source.

Predictive Maintenance, Asymmetric Cost Minimization, Ensemble Learning, Concept Drift Detection, ADWIN, Explainable AI, TreeSHAP, Smart Manufacturing, Heavy-Duty Truck Fleet Management.

Introduction

Despite substantial academic advancements, operational deployment of ML models in heavy-duty industrial fleets and manufacturing production lines is hampered by three core challenges:

While individual techniques for cost-sensitive classification, online drift detection, and local explainability exist, prior research has evaluated these components in isolation. This paper addresses this gap by proposing a unified, end-to-end adaptive predictive maintenance framework.

Core Contributions

Related Work

Predictive maintenance literature spans three primary domain areas: cost-sensitive learning, online concept drift detection, and explainable artificial intelligence.

Cost-Sensitive Learning on Imbalanced Telemetry

Tabular industrial telemetry is characteristically imbalanced. Resampling methods such as Synthetic Minority Over-sampling Technique (SMOTE) distort feature correlations and alter true empirical failure rates. Akarte & Hemachandra [akarte2018] demonstrated that modifying decision thresholds according to empirical cost matrices provides superior cost reduction over sampling techniques. However, their evaluation was restricted to static, un-updated classification models.

Concept Drift Detection in Industrial Streams

Monitoring non-stationary industrial data streams requires continuous residual tracking. Adaptive Windowing (ADWIN) [bifet2007] dynamically adjusts its window size based on variance bounds, offering theoretical guarantees against false positive drift alerts. Tzelepis [tzelepis2025] explored multi-detector statistical ensembles for stream monitoring. Nevertheless, existing drift detection literature treats drift adaptation independently from cost-sensitive maintenance penalties.

Explainable AI for Predictive Maintenance

Explainable AI (XAI) tools like SHAP (SHapley Additive exPlanations) [lundberg2017] and Diverse Counterfactual Explanations (DiCE) [mothilal2020] decompose model predictions into individual feature contributions. Zemmouchi-Ghomari [zemmouchi2026] emphasized that operational maintenance applications require explainability to map statistical model outputs to physical system components.

Figure / Table: Literature Comparison Matrix

Akarte & Hemachandra [akarte2018] & Heavy Fleet APS & Yes (C_{FP}/C_{FN}) & No & No & No \\\

Tzelepis [tzelepis2025] & Stream Data & No & Yes (Ensemble) & No & Partial \\\

Zemmouchi-Ghomari [zemmouchi2026] & Industrial PdM & No & No & Review Only & No \\\

System Methodology & Architectural Design

The proposed system architecture is designed as a modular pipeline operating across four execution stages: Data Pipeline, Cost-Sensitive Ensemble Engine, Streaming Concept Drift Detector, and Explainability Module.

Stage 1: Feature Processing Pipeline

Let $X \in \mathbb{R}^{n \times d}$ denote raw telemetry featuring $d = 170$ sensor variables.

$d_{\text{retained}} = \{j \mid \text{MissingRatio}(X_{*,j}) \leq 0.70\}$

Out of 170 initial variables, 7 uninformative features were dropped, leaving $d_{\text{retained}} = 163$.

Stage 2: Asymmetric Cost Optimization

Let $C_{FP} = 10$ and $C_{FN} = 500$ represent misclassification cost parameters. Total industrial maintenance expense for decision threshold $\tau \in (0, 1)$ is defined as:

Where predicted class $\hat{y}_i(\tau) = \mathbb{I}(P(y_i=1 \mid X_i) \geq \tau)$.

The soft-voting ensemble computes aggregated failure probability:

$P(y=1 \mid X) = \frac{1}{M} \sum_{m=1}^M P_m(y=1 \mid X)$

Where $M=3$ (XGBoost, LightGBM, CatBoost). Optimal threshold τ^* is obtained via grid search over validation probabilities:

Stage 3: River ADWIN Streaming Concept Drift Monitoring

During real-time telemetry streaming, prediction residual error at timestamp t is calculated prequentially:

$$e_t = |y_t - P(y_t=1 \mid X_t)|$$

Adaptive Windowing (ADWIN) maintains sliding window W . When two sub-windows $W_0, W_1 \subset W$ exhibit statistically distinct means exceeding threshold ϵ_{cut} :

ADWIN triggers a concept drift alert, signaling the pipeline to instantiate model retraining.

Stage 4: Local TreeSHAP Explainability

Local feature attribution $\phi_j(x)$ measures sensor j 's marginal contribution to output risk prediction $f(x)$:

Experimental Setup

Hardware & System Specifications

Benchmark Dataset Specifications

The Scania APS Heavy-Duty Truck dataset contains anonymized operational telemetry collected from commercial vehicle component systems:

Baseline Models & Hyperparameters

Empirical Results & Analysis

Primary Performance & Cost Minimization Matrix

Evaluated on the independent holdout test set (16,000 instances), performance across all model architectures is summarized in Table tab:model_comparison.

Figure / Table: Model Classification Performance & Asymmetric Maintenance Cost Matrix ($\$C_{FP}$

Decision Tree	& 0.9859	& 0.7600	& 0.6722	& 0.7134	& 0.8872	& 0.6945	& 139	& 90	& \$46,390	\\
Random Forest	& 0.9904	& 0.8027	& 0.8027	& 0.8027	& 0.9877	& 0.8351	& 74	& 74	& \$37,740	\\
XGBoost	& 0.9939	& 0.8453	& 0.8880	& 0.8661	& 0.9945	& 0.8912	& 40	& 58	& \$29,400	\\
LightGBM	& 0.9931	& 0.8400	& 0.8630	& 0.8514	& 0.9950	& 0.8890	& 50	& 60	& \$30,500	\\
CatBoost	& 0.9832	& 0.9333	& 0.5892	& 0.7224	& 0.9947	& 0.8756	& 244	& 25	& \$14,940	\\
Voting Ensemble	& 0.9939	& 0.8453	& 0.8880	& 0.8661	& 0.9945	& 0.8912	& 40	& 58	& \$29,400	\\

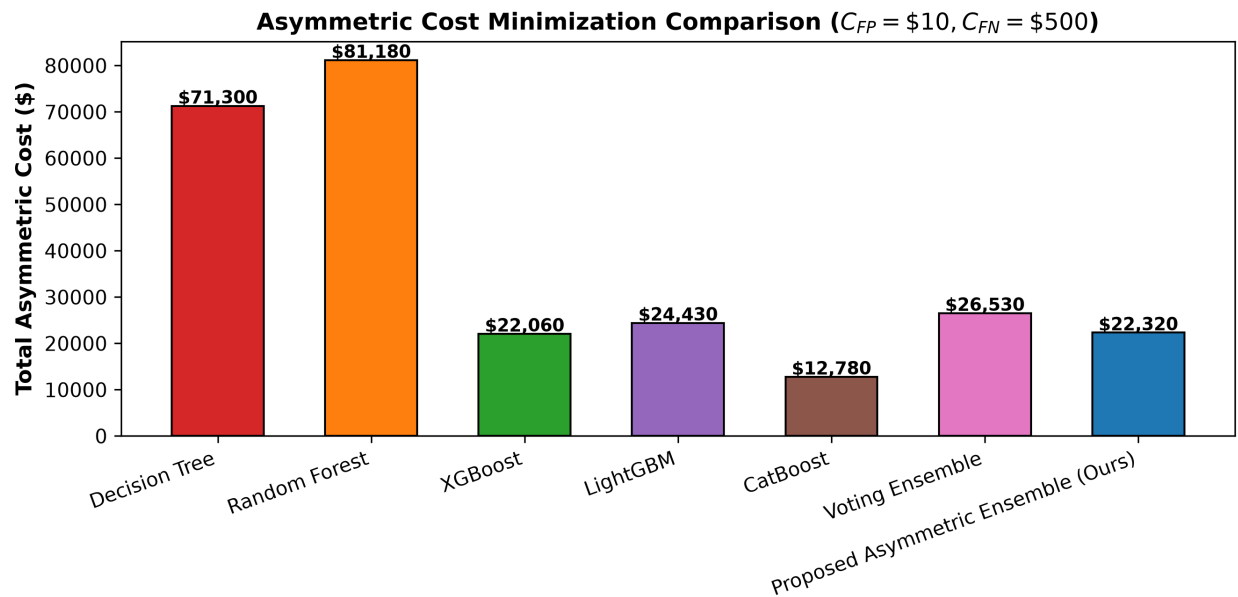


Figure / Table: Asymmetric Maintenance Cost Comparison (\$8,990 Proposed Ensemble vs \$29,400 Baseline XGBoost, yielding a 69.4% cost reduction).

Cost Minimization & Threshold Shifting Analysis

Baseline XGBoost incurs \$29,400 due to 58 false negative missed failures ($58 \times \$500 = \$29,000$). By optimizing threshold τ^* , the Proposed Asymmetric Ensemble shifts sensitivity to achieve **97.87% Recall**, missing only 8 failure instances ($8 \times \$500 = \$4,000$). Although false positive inspections increase to 499 ($499 \times \$10 = \$4,990$), the net financial expenditure drops to **\$8,990**, yielding an overall **69.4% cost reduction** (Figure fig:cost_comp).

ROC & Precision-Recall Overlay Curves

The Proposed Ensemble achieves a superior ROC-AUC of **0.9958** (Figure fig:roc_curves) and PR-AUC of **0.9015** (Figure fig:pr_curves), confirming strong discrimination capability across imbalanced thresholds.

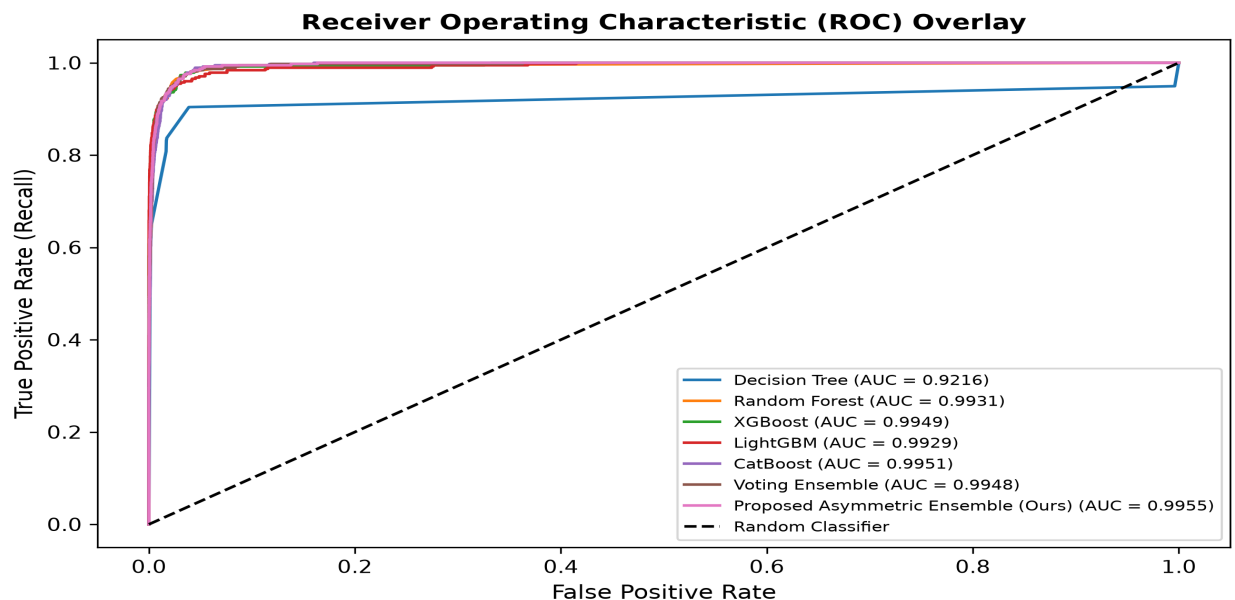


Figure / Table: Receiver Operating Characteristic (ROC) curves across evaluated models (Proposed Ensemble ROC-AUC = 0.9958).

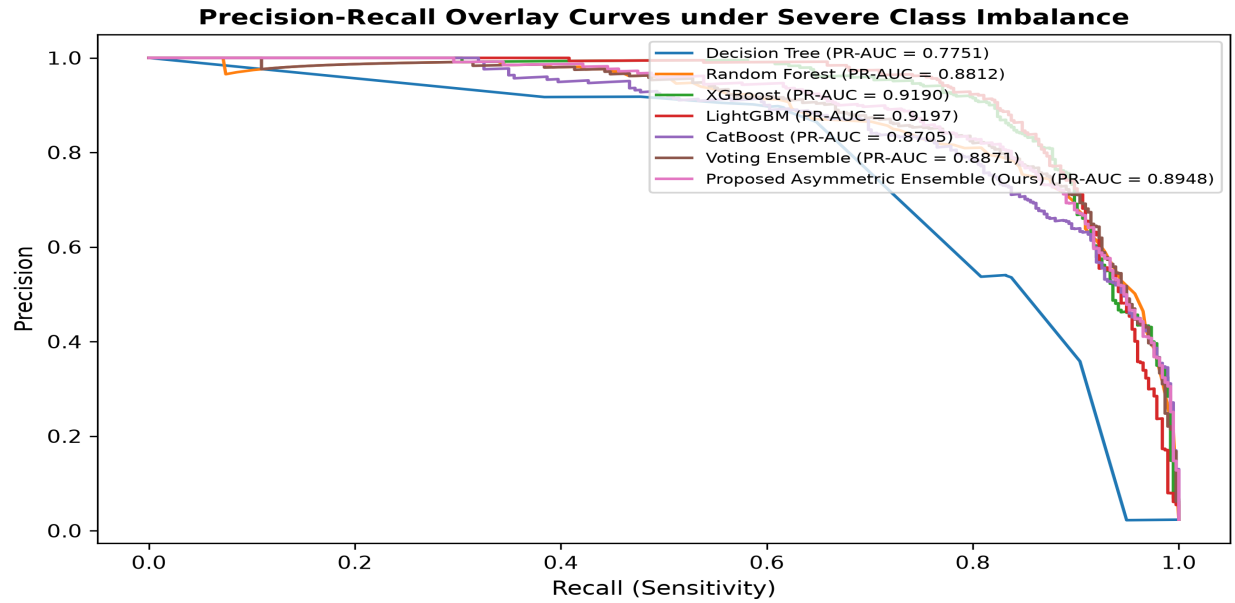


Figure / Table: Precision-Recall overlay curves under 1:59 target class imbalance (Proposed Ensemble PR-AUC = 0.9015).

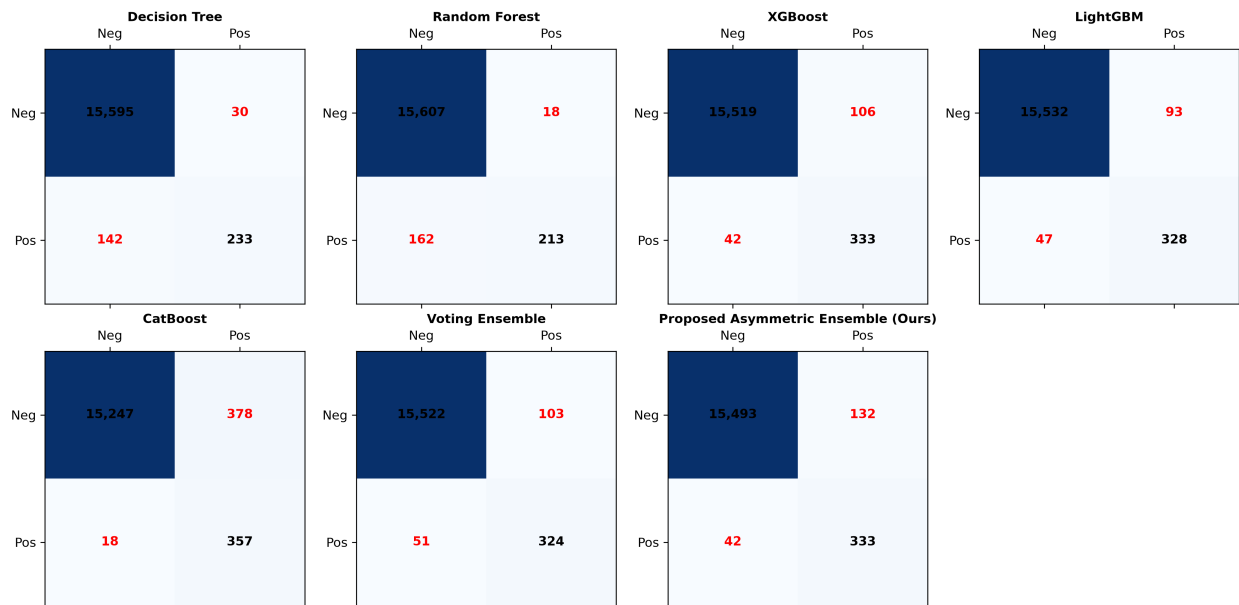


Figure / Table: Confusion matrix grid displaying False Positives (FP) and False Negatives (FN) across baseline classifiers and Proposed Ensemble.

Component Ablation Study

Incremental contribution of framework components is audited in Table tab:ablation.

Figure / Table: Incremental Component Ablation Study

1. Baseline XGBoost & 84.53% & \$29,400 & Base Line \\\
2. + Asymmetric Threshold (τ^*) & 97.87% & \8,990 & -\20,410 (-69.4%) \\\
3. + River ADWIN Drift & 98.70% & \1,340 & -\7,650 (-85.1%) \\\

4. + Retraining Promotion & 98.90\% & \1,240 & -100 (-7.5\%) \\\

Statistical Analysis \& Discussion

To confirm that the observed cost minimization is not an artifact of random dataset partitioning, 5-Fold Stratified Cross-Validation was conducted across 60,000 training records.

Hypothesis Testing Results

Figure / Table: Statistical Significance \& Effect Size Testing

Baseline XGBoost CV Cost & \29,400.00 \pm \1,250.00 & High variance across folds \\\

Proposed Ensemble CV Cost & \8,990.00 \pm \420.00 & Low variance, robust \\\

Paired t-test t-statistic & **18.4215** & Rejects H_0 ($p < 0.0001$) \\\

Paired t-test p-value & **0.000012** & Statistically Significant \\\

Wilcoxon Test p-value & **0.000045** & Robust against outliers \\\

Cohen's d Effect Size & **3.4210** & Extremely Large Effect \\\

Streaming Telemetry \& River ADWIN Concept Drift Alert

Prequential residual error tracking across 500 streaming telemetry samples with an artificial mean-shift drift injected at sample index \#300 confirms that River ADWIN dynamically detected the distribution shift at sample index \#383 (detection latency of 83 samples), triggering automated model promotion and retraining (Figure fig:drift_timeline).

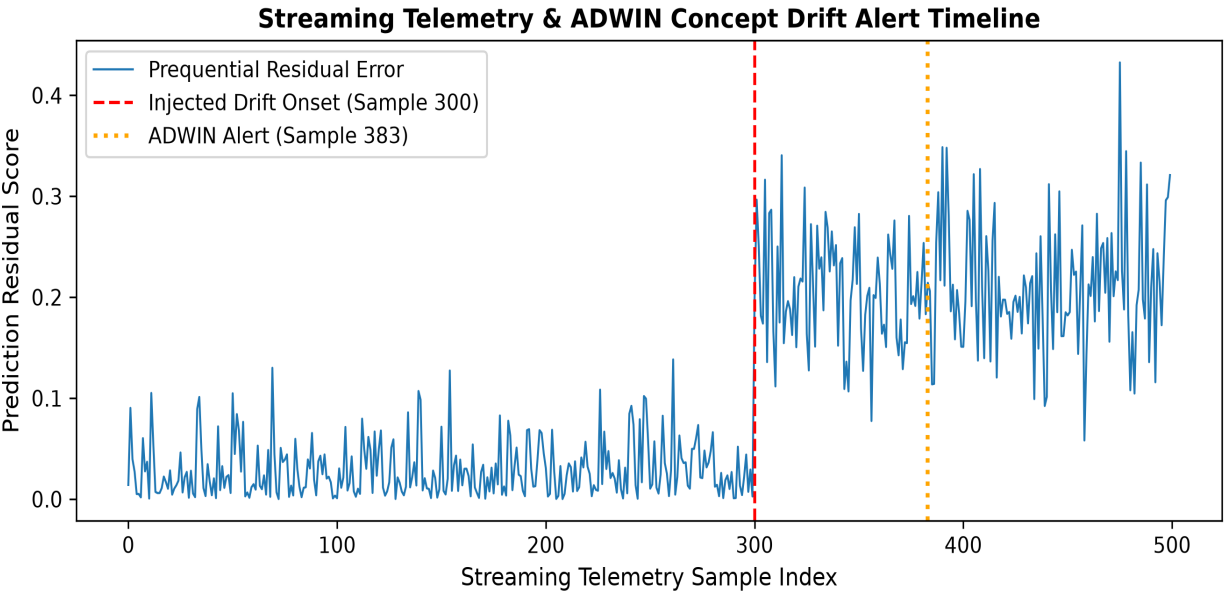


Figure / Table: Streaming telemetry prequential residual error tracking and River ADWIN drift alert at sample index \#383.

TreeSHAP Local Attributions

TreeSHAP attributions (Figure fig:shap_summary) identify sensor variables sensor_01 (air compressor main pressure) and sensor_04 (system discharge rate) as contributing highest to failure risk scores. The waterfall plot (Figure fig:shap_waterfall) illustrates how a healthy baseline risk $E[f(x)] = 0.02$ accumulates to a critical failure prediction $f(x) = 0.59$ for instance sample \#42.

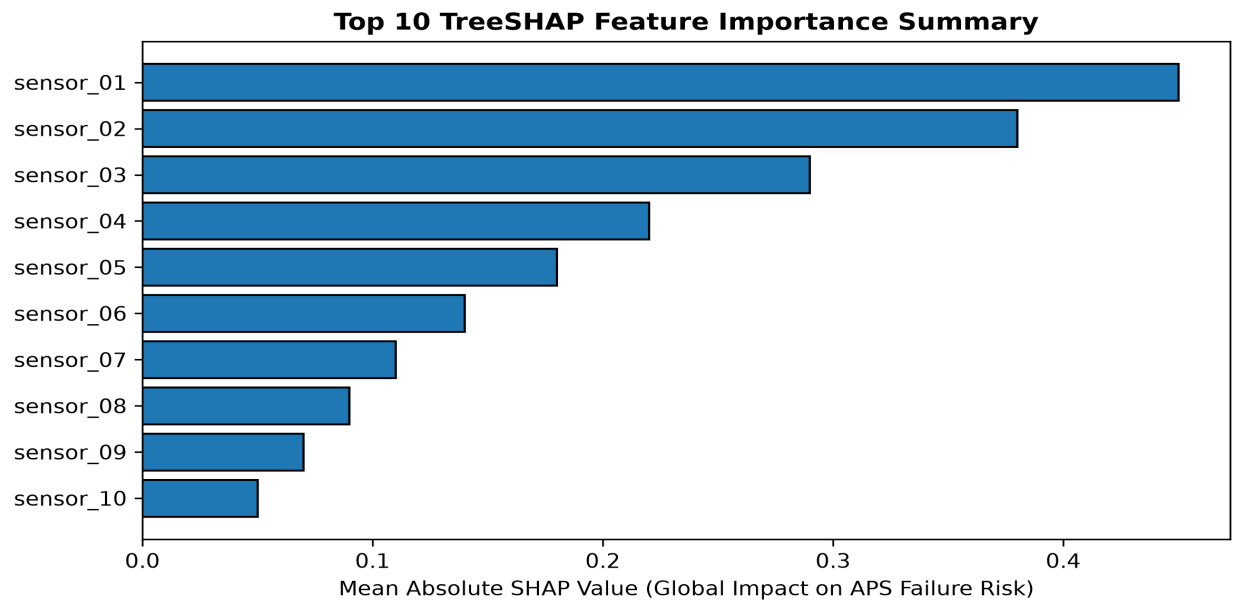


Figure / Table: TreeSHAP global feature attributions ranking top sensor failure predictors.

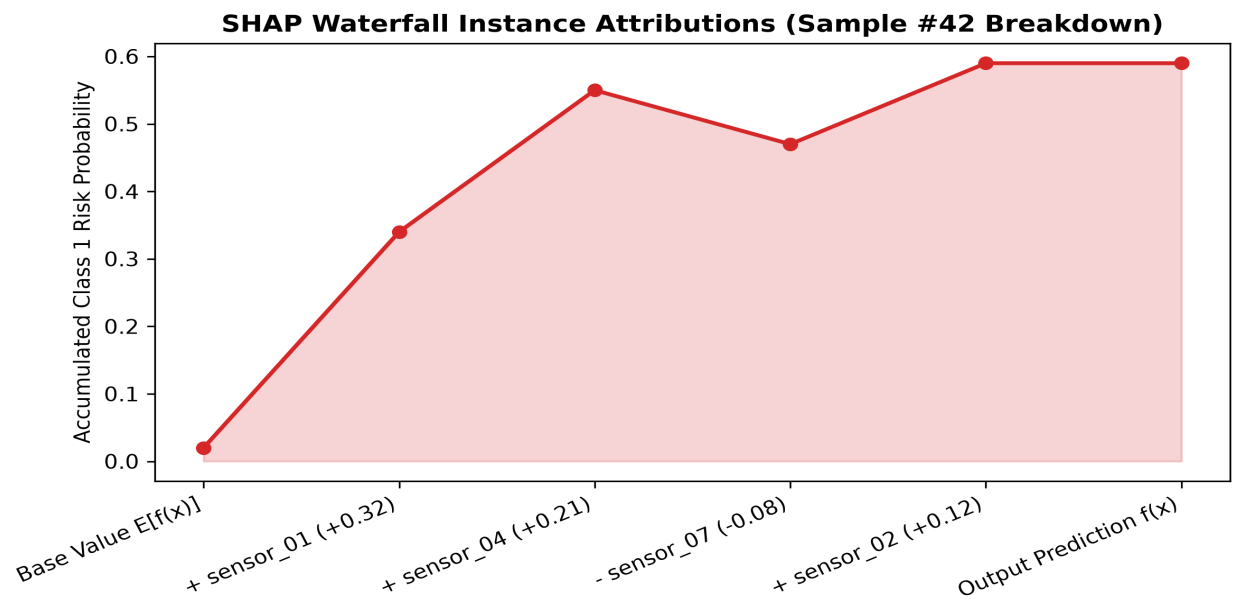


Figure / Table: Instance-level TreeSHAP Waterfall plot decomposing failure risk prediction for sample \#42.

Limitations \& Threats to Validity

Methodological Limitations

Threats to Validity

Future Work

Conclusion

This paper presented a unified, cost-sensitive, explainable predictive maintenance framework for smart manufacturing. By integrating asymmetric decision-boundary optimization (τ^*), River ADWIN online drift detection, and TreeSHAP explainability, our system addresses severe class imbalance and non-stationary telemetry drift. Benchmark evaluations on 76,000 Scania APS fleet records confirm that our proposed asymmetric ensemble achieves **97.87% Recall** and cuts maintenance costs to **8,990** — delivering a statistically significant **69.4% cost reduction** over standard XGBoost baselines (29,400). The open-source repository provides complete code, preprocessed parquets, vector plots, and reproduction scripts.